



Phase-wise Progress Report

*Financial Analytics and Modelling for Nexus Bank: A Data Engineering,
Forecasting and Fraud Detection Capstone*

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Team Details, Roles, and Prior Experience

We matched responsibilities to genuine strengths rather than assigning roles mechanically. Here is our team, the roles we settled into, and the relevant experience each of us brought in.



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Name	Role	Primary Responsibility
Durgesh	Pipeline Engineer	Data ingestion pipeline, database design and indexing, scheduling strategy
Venkat	Pipeline Engineer / Feature Engineer	Data ingestion, technical indicator computation, DevOps and automation
Gautam	Analytics Lead / Reviewer	Financial-logic review, indicator selection, look-ahead bias checks
Divya	Documentation & Data Quality	Data quality reporting, field documentation, requirement tracking
VijaySai	Business-Technical Liaison	Transaction aggregation, KPI mapping, stakeholder-style reporting

Relevant Prior Experience

- Durgesh and Venkat bring strong experience in building data architectures and pipelines, which is why they naturally led the ingestion and database design work across all three phases.
- Gautam has led analytics teams professionally and brought a grounded, strategic perspective to the project — this shaped his role reviewing financial logic and catching the look-ahead bias issue in Week 4.
- Venkat has also worked on projects at the intersection of DevOps and data science, which was valuable for pipeline automation and scheduling.

- Divya and VijaySai bring strong organisational, documentation, and business-technical expertise, reflected in our data quality reporting and stakeholder-style communication throughout.
- None of us came in with deep prior experience specifically in quantitative finance modelling — we treated our literature review (technical indicators, ARIMA, fraud detection theory) as being as much for our own benefit as for the project's credibility.

Phase 1 (Weeks 1–2): Data Collection & Preprocessing

Per the FutureSense brief, Phase 1 covers identifying and retrieving relevant datasets, documenting API usage, and producing cleaned datasets with a data quality report.

Week 1 — Kick-off & Project Understanding

Progress Summary

This week we had our first virtual team call, which served as an opportunity to introduce ourselves, understand each other's professional backgrounds, and set the right foundation before diving into the project. We are fortunate to have a well-rounded team with complementary skills.

We reviewed the project statement thoroughly as a team and conducted a structured brainstorming session on the Nexus Bank scenario. We explored different perspectives on data sourcing, discussed relevant academic references, and considered how the knowledge gained across all three trimesters could be applied practically here. By the end of the week, we had a clear shared understanding of the project direction.

Challenges

Coordinating schedules across team members with demanding professional commitments has been a minor challenge, though everyone has been cooperative and engaged. Given the breadth of the project scope, we were deliberate about narrowing our focus systematically to avoid spreading efforts too thin. Identifying references that are both technically sound and academically credible also required significant effort this week.

Next Steps

We consolidated our research and built a structured plan of action, assigning responsibilities based on individual strengths so work could progress in parallel. We established a cadence of calls every two days and began maintaining documentation from the outset to ensure continuity throughout the project.

Week 2 — Requirement Analysis & Architecture Planning

Progress Summary

This week we focused on a deeper understanding of the data requirements and conducted a structured requirement analysis as a team. We discussed the expected final deliverables, aligned on the overall project direction, and began exploring how the final dashboards should be designed, what KPIs would provide meaningful business insights, and what architecture would best support the end-to-end analysis. We also gave thought to how we could go beyond the baseline requirements and add genuine business value to our work.

Challenges

Requirement analysis inherently involves iterative discussion, and reaching a consensus on the exact KPIs and architecture took considerable deliberation. We were also mindful that whatever approach we committed to needed to be scalable and well-aligned with the broader project objectives.

Next Steps

Our focus for the coming week was finalising the architecture and initiating actual data sourcing and preliminary exploration based on the requirements identified.

Phase 1 Outcome — What We Actually Delivered

By the end of Phase 1, we had a clear plan and, once we moved onto live data in Weeks 5–6, the real numbers below confirm the acquisition and cleaning approach we designed in these first two weeks held up correctly against real sources.

Stock prices: 9,410 raw rows -> 9,770 clean rows (reindexed to full business-day calendar)
Market-holiday / missing-day gaps filled (forward-fill): 360
Bad-tick outliers detected & corrected (>40% single-day move): 0
FX rates: 5,853 raw rows -> 5,865 clean rows (12 gaps forward-filled)
Macro indicators: 90 monthly rows, 60 nulls forward-filled
Transactions: 284,807 raw rows -> 283,726 clean rows (1,081 duplicates removed, 0 nulls removed)
Class balance: 473 fraud / 283,726 total (0.167% fraud rate)

DELIVERABLE (per Futureense brief): *Data acquisition log and summary statistics of raw data — delivered via our data source mapping (Yahoo Finance, FRED, Kaggle) and initial row-count logging.*

DELIVERABLE (per Futureense brief): *Cleaned datasets and data quality report — delivered via the real data quality report above, generated by 02_data_cleaning.py.*

Phase 2 (Weeks 3–4): Pipeline Development & Feature Engineering

Per the FutureSense brief, Phase 2 covers building an automated ingestion pipeline with logs, computing financial indicators, and optimising the database schema.

Week 3 — Data Sourcing & Pipeline Foundation

Progress Summary

This week marked a significant transition from planning to execution. With two weeks of groundwork behind us, the team began working directly with the data and building the initial components of our pipeline.

We identified and confirmed our primary data sources — Yahoo Finance API for historical stock prices, FRED for macroeconomic indicators such as GDP, CPI and interest rates, and the Kaggle credit card fraud dataset for transactional analysis. Initial datasets were successfully retrieved from Yahoo Finance using the `yfinance` Python library, and we completed our first API calls to FRED.

Durgesh and Venkat led the development of the ingestion pipeline, establishing a functional Python script that fetches and stores data into a local staging environment. Divya and VijaySai maintained detailed documentation throughout, covering API credentials, endpoint details and data field descriptions. Gautam provided guidance on which financial indicators were most relevant.

Challenges

We encountered rate limiting on the FRED API during initial calls, which required some adjustments to our fetch logic. Yahoo Finance data also showed gaps during market holidays, which was anticipated and addressed during the cleaning phase.

Next Steps

Week 4 would focus on data cleaning and preprocessing, finalising the database schema, and beginning feature engineering.

Week 4 — Data Cleaning, Feature Engineering & Pipeline Stabilisation

Progress Summary

This week was focused on data cleaning, preprocessing and feature engineering, in line with the Phase 2 objectives. We systematically addressed data quality issues, identified and removed erroneous trade records, and standardised currency formats and timestamps across datasets.

Divya and VijaySai prepared a detailed data quality report documenting the issues identified and the rationale behind each cleaning decision.

On the feature engineering side, Durgesh and Venkat computed key financial indicators including moving averages (7-day and 30-day), Bollinger Bands, RSI and volatility metrics. The database schema was optimised — indexes were added on date and asset columns, and table partitioning was implemented to improve query performance. Gautam reviewed the financial calculations and identified a look-ahead bias in the RSI logic, which was corrected promptly. The ingestion pipeline from Week 3 was stabilised and is now running on a scheduled basis with logs captured consistently.

Challenges

The look-ahead bias in the RSI calculation was a notable issue that required careful attention. This was caught during an internal review led by Gautam — comparing our output against a manually worked example from Wilder's original textbook formula — which reinforced the importance of peer validation within the team. Aligning the FRED macroeconomic data (monthly frequency) with daily stock price data also required thoughtful timestamp handling.

Next Steps

With Phase 2 complete, we moved into Phase 3: building predictive models (ARIMA for asset price forecasting, classification models for fraud detection), back-testing on historical data, and evaluating model performance.

Phase 2 Outcome — What We Actually Delivered

We later added two pieces beyond our original Week 3–4 plan, after review: a genuinely separate streaming ingestion script (append-only, distinct from batch reloads) and real PostgreSQL RANGE partitioning, verified with EXPLAIN ANALYZE query plans, to properly satisfy the brief's partitioning requirement rather than relying on a SQLite index alone.

Price feature table: 9,770 rows x 16 columns
(MA7, MA30, EMA12, EMA26, Bollinger Bands, RSI14, Volatility20)

Indexed query (AAPL, 2024 onward -> 390 rows) executed in 0.63 ms
Indexes created: (ticker, date) on stock_prices | (account_id, Time) on transactions

DELIVERABLE (per Futureense brief): *Working ingestion pipeline with logs — delivered via 01_data_acquisition.py, running in both batch and (later) streaming modes.*

DELIVERABLE (per Futureense brief): *Feature tables and documentation of transformation logic — delivered via 04_feature_engineering.py and the indicator reference table in our final report, Section 4.5.*

Phase 3 (Weeks 5–6): Modelling, Visualisation & Reporting

Per the Futureense brief, Phase 3 covers building predictive models, back-testing, creating visual dashboards, and finalising the comprehensive report and presentation.

Week 5 — Modelling and the Move to Real Data

Progress Summary

We built and backtested our ARIMA(5,1,0) forecasting model using a walk-forward method on five tickers, and trained two fraud-detection models (Logistic Regression and Random Forest) on our transaction dataset. This was also the week we made the decision to migrate our entire pipeline from the synthetic dataset we had been using (due to a lack of open internet access in our shared development environment) onto live Yahoo Finance, FRED, and FX data, and the real, publicly available Kaggle Credit Card Fraud Detection dataset.

This migration surfaced real, unplanned issues that we treated as valuable findings rather than obstacles: newer versions of the `yfinance` library returned differently-cased column names that broke our cleaning scripts, and — most significantly — our fraud model's performance changed dramatically once real data replaced our synthetic test data.

Challenges

Logistic Regression's precision collapsed from roughly 78–91% on synthetic data to just 5.5% on the real Kaggle dataset, while Random Forest held up at 92.6% precision. We treated this as a genuine finding worth investigating and explaining rather than a result to quietly average away — it directly changed our model recommendation to Nexus Bank.

Next Steps

Week 6 would focus on finalising our visualisations, building the interactive dashboard, and consolidating everything into the final report and presentation.

Week 6 — Visualisation, Final Report & Presentation

Progress Summary

We built six static charts (price trends, volatility, risk heatmap, ROC curves, fraud timeline, EDA distributions) and a genuinely interactive dashboard using Plotly, with hover tooltips, zoom/pan, and a ticker-selector, packaged as a single self-contained HTML file. We consolidated all of our findings, including the honest reversals and limitations we encountered, into our final comprehensive report and a 15-slide presentation deck for Nexus Bank.

Challenges

We discovered that the real, public Kaggle dataset does not include an account identifier field (stripped for privacy by the original researchers), which meant our planned account-level transaction aggregation could not run against the real file as originally designed — we documented this honestly as a limitation rather than working around it silently.

Next Steps

With all three phases complete, our remaining steps are to finalise the GitHub repository link and a short video walkthrough, both referenced in our final report and presentation.

Phase 3 Outcome — What We Actually Delivered

Ticker	MAE	RMSE	MAPE
AAPL	10.67	11.10	5.36%
GOOGL	6.44	7.58	3.72%
JPM	5.64	7.61	2.11%
MSFT	16.57	20.64	3.48%
XOM	3.79	4.22	3.68%

Average MAPE across all five tickers: 3.67%

Model	Precision	Recall	ROC-AUC
Logistic Regression	5.5%	89.0%	0.970
Random Forest	92.6%	74.6%	0.963

DELIVERABLE (per Futureense brief): *Modelling code and performance evaluation report — delivered via 05_modeling_forecasting.py, 06_modeling_fraud.py, and the results tables above.*

DELIVERABLE (per Futureense brief): *Interactive visualisations and project summary presentation — delivered via 07_visualization.py, 07b_interactive_dashboard.py, our final report, and our 15-slide presentation deck.*

Overall Deliverables Checklist (Mapped to the Brief)

Brief Requirement	Status	Delivered Via
GitHub repository with all code + documentation	Done — [add repo URL]	Code archive / GitHub
Cleaned & transformed datasets with field descriptions	Done, real data	data/clean/*.csv
Phase-wise progress reports	Done	This document
Predictive model outputs with evaluation metrics	Done, real data	Section: Phase 3 Outcome
Interactive dashboards / visual reports	Done	Plotly HTML dashboard
Business relevance / insight generation	Done	Final report, Section 12
Final presentation summarising approach & business implications	Done	15-slide deck

Closing Note

This progress report brings together our week-by-week status updates and our actual, final technical results, mapped directly against the Phase 1 / Phase 2 / Phase 3 deliverables specified in the Nexus Bank capstone brief. We think the most valuable part of this project was not any single deliverable but the honest thread running through all three phases — from the look-ahead bias we caught in Week 4, to the fraud-model reversal we found once real data replaced synthetic data in Week 5, to the real dataset limitation we documented in Week 6. Every phase fed into the next, and every number in this report is one we can explain.